

DATA ANALYTICS WITH COGNOS - GROUP 5

PROJECT: CUSTOMER CHURN PREDICTION

PHASE 3: DEVELOPMENT PART 1

TEAM MEMBER

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PHASE 3 - DOCUMENT SUBMISSION

**TOPIC: IN THIS SECTION BEGIN BUILDING PROJECT BY
LOADING AND PREPROCESSING THE DATASET**



CUSTOMER CHURN PREDICTION

INTRODUCTION

In today's highly competitive business landscape, retaining customers is of paramount importance. Customer churn, the rate at which customers discontinue their relationship with a company, can significantly impact a company's profitability and long-term success. To address this challenge, this project focuses on customer churn prediction. By leveraging data analytics and machine learning techniques, we aim to develop a predictive model that can identify customers at risk of churning. This model will enable businesses to take proactive measures to retain valuable customers, ultimately increasing customer loyalty and sustaining growth. In this project, we will explore the data, develop predictive models, and provide actionable insights to reduce churn and enhance customer satisfaction.

As we embark on the development phase of our customer churn prediction project, we move closer to realizing the goal of building a robust and data-driven solution to address one of the most pressing challenges in the business world. The development phase represents a critical juncture in our project journey. During this phase, we will leverage cutting-edge data analytics and machine learning techniques to transform raw data into actionable insights. Our mission is to create a predictive model capable of identifying customers on the brink of churning, empowering businesses to implement timely interventions. This phase involves data preprocessing, feature engineering, model selection, and rigorous testing to ensure the accuracy and effectiveness of our churn prediction system. As we delve deeper into the development process, we are one step closer to delivering a tool that will not only reduce churn rates but also foster lasting customer relationships, driving business success in an increasingly competitive market.

DATASET

Telco customer churn:

The dataset contains 7043 rows and 21 features. Each row represents a customer, each column contains the customer's attributes described in the column Metadata

Dataset Link: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

customer	gender	SeniorCiti Partner	Depender tenure	PhoneSer	MultipleLi	InternetS	OnlineSec	OnlineBac	DevicePrc	TechSupp	Streaming	Streaming	Contract	Paperless	PaymentM	MonthlyC	TotalChar	Churr
7590-VHV	Female	0 Yes	No	1 No	No phone	DSL	No	Yes	No	No	No	No	Month-to	Yes	Electronic	29.85	29.85	No
5575-GNV	Male	0 No	No	34 Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	Mailed ch	56.95	1889.5	No
3668-QPYI	Male	0 No	No	2 Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to	Yes	Mailed ch	53.85	108.15	Yes
7795-CFO	Male	0 No	No	45 No	No phone	DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank trans	42.3	1840.75	No
9237-HQIT	Female	0 No	No	2 Yes	No	Fiber opti	No	No	No	No	No	No	Month-to	Yes	Electronic	70.7	151.65	Yes
9305-CD5I	Female	0 No	No	8 Yes	Yes	Fiber opti	No	No	Yes	No	Yes	Yes	Month-to	Yes	Electronic	99.65	820.5	Yes
1452-KIOV	Male	0 No	Yes	22 Yes	Yes	Fiber opti	No	Yes	No	No	Yes	No	Month-to	Yes	Credit car	89.1	1949.4	No
6713-OKO	Female	0 No	No	10 No	No phone	DSL	Yes	No	No	No	No	No	Month-to	No	Mailed ch	29.75	301.9	No
7892-POO	Female	0 Yes	No	28 Yes	Yes	Fiber opti	No	No	Yes	Yes	Yes	Yes	Month-to	Yes	Electronic	104.8	3046.05	Yes
6388-TABK	Male	0 No	Yes	62 Yes	No	DSL	Yes	Yes	No	No	No	No	One year	No	Bank tran	56.15	3487.95	No
9763-GR5I	Male	0 Yes	Yes	13 Yes	No	DSL	Yes	No	No	No	No	No	Month-to	Yes	Mailed ch	49.95	587.45	No
7469-LKBC	Male	0 No	No	16 Yes	No	No	No intern	No intern	No intern	No intern	No intern	No intern	Two year	No	Credit car	18.95	326.8	No
8091-TTVZ	Male	0 Yes	No	58 Yes	Yes	Fiber opti	No	No	Yes	No	Yes	Yes	One year	No	Credit car	100.35	5681.1	No
0280-XJGE	Male	0 No	No	49 Yes	Yes	Fiber opti	No	Yes	Yes	No	Yes	Yes	Month-to	Yes	Bank tran	103.7	5036.3	Yes
5129-JLPI	Male	0 No	No	25 Yes	No	Fiber opti	Yes	No	Yes	Yes	Yes	Yes	Month-to	Yes	Electronic	105.5	2686.05	No
3655-SNQ	Female	0 Yes	Yes	69 Yes	Yes	Fiber opti	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No	Credit car	113.25	7895.15	No
8191-XWS	Female	0 No	No	52 Yes	No	No	No intern	No intern	No intern	No intern	No intern	No intern	One year	No	Mailed ch	20.65	1022.95	No
9959-WOF	Male	0 No	Yes	71 Yes	Yes	Fiber opti	Yes	No	Yes	No	Yes	Yes	Two year	No	Bank tran	106.7	7382.25	No
4190-MFLI	Female	0 Yes	Yes	10 Yes	No	DSL	No	No	Yes	Yes	No	No	Month-to	No	Credit car	55.2	528.35	Yes
4183-MYFI	Female	0 No	No	21 Yes	No	Fiber opti	No	Yes	Yes	No	No	Yes	Month-to	Yes	Electronic	90.05	1862.9	No

NECESSARY STEPS TO FOLLOW

1.Import Libraries:

Start by importing necessary libraries

Program

```
From mpl_toolkits.mplot3d import Axes3D
```

```
From sklearn.preprocessing
```

```
import StandardScaler
```

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
import pandas as pd
```

2.Load The Dataset:

Load your dataset into a pandas dataframe.You can typically find customer churn datasets in csv format., but you can adapt this code to other formats as needed.

Program

```
Print(os.listdir(['WA_Fn-UseC_-Telco-Customer-Churn.csv']))
```

3.Logistic Regression:

Perform logistic prediction for prediction and classification. Logistic regression models can help teams identify data anomalies, which are predictive of fraud.

Program

```
Def logistic_regression(frame, grid):  
  
Logit = Model(LogisticRegression, frame, metrics)  
  
Print_result(logit.grid_search('roc_auc', grid), "cross-validation")  
Print_result(logit.train_and_evaluate_test(), "test set")  
  
Print("Best hyperparameters:", logit.hyperparameters)  
  
Logistic_regression(df_processed, {'C': np.logspace(-4, 4, 100, base=10)})
```

4.Support Vector Machine:

SVM is used to classify data by finding the optimal decision boundary that maimally seperates different classes. It aims to find the best hyperplane that maximizes the margin between support vectors, enabling effective classification even in complex, non linear scenarios.

Program

```
Def svc_rbf(frame, grid):  
  
Rbf = Model(SVC, frame, metrics, fixed_params= {'kernel': 'rbf', 'probability': True})  
Print_result(rbf.grid_search('roc_auc', grid, num_folds=4), "cross-validation")  
Print_result(rbf.train_and_evaluate_test(), "test set")  
  
Print("Best hyperparameters:", rbf.hyperparameters)  
  
Grid_rbf = {'C': np.logspace(-4, 1, 10, base=2), 'gamma': np.logspace(-6, 2, 10, base=2)}  
Svc_rbf(df_processed, grid_rbf)
```

5. Nearest Neighbour Classifier

It aims at labelling previously unseen query objects while distinguishing two or more destination classes.

Program

```
Def knn_clf(frame, grid):

Knn = Model(KNeighborsClassifier, frame, metrics)

Print_result(knn.grid_search('roc_auc', grid), "cross-validation")
Print_result(knn.train_and_evaluate_test(), "test set")

Print("Best hyperparameters:", knn.hyperparameters)

Knn_clf(df_processed, {'n_neighbors': [i for i in range(10, 50, 2)]})
```

6. Feature Engineering:

Depending on your dataset, you may need to create new features or transform existing ones. This can involve one-hot encoding categorical variables, handling data/time, data, or scaling numerical features.

Program

```
# Remove Gender

Features = ['gender']

Df_processed.drop(features, axis=1, inplace=True)

Logit = Model(LogisticRegression, df_processed, metrics)

Print_result(logit.grid_search('roc_auc', logit_grid), "cross-validation")
```

7. Feature Scaling:

Apply feature scaling to normalize your data, ensuring that all features have similar scales. Standardization is a common choice.

Program

For feat in numerical_cols: distplot(feats, df)

IMPORTANCE OF LOADING AND PREPROCESSING DATASET

Loading and preprocessing the dataset is an important first step in building any machine learning model. However, it is especially important for customer churn prediction models, as customer prediction datasets are often complex. By loading and preprocessing the dataset, we can ensure that the machine learning algorithm is able to learn from the data effectively and accurately.

CHALLENGE INVOLVED IN LOADING AND PREPROCESSING CUSTOMER CHURN PREDICTION DATASET

There are a number of challenges involved in loading and preprocessing a customer prediction dataset, including:

- **Features And Exploratory Analysis:**

Businesses face several roadblocks and churn risks at this stage of building predictive churn models such as lack of information, target leakage, and the need for optimal feature transformations. Along with domain knowledge, businesses must also have the required skills and creativity to build robust predictive churn models. Therefore, it is important that companies execute careful exploratory analysis and build auxiliary models before embarking on building an overall churn prediction model. Exploratory analysis can also help in revealing reciprocity, irregularities, outliers, and, relationships between different functions, which wouldn't be possible with domain knowledge alone.

- **Validating Churn Model Performance:**

For accurate customer churn prediction, it's essential to choose the correct metric to optimize and validate datasets. The precision of a churn model not only impacts performance but also affects decision-making. As such, businesses need to employ different strategies to validate the performance of a churn model prior to its implementation. Also, businesses need to monitor several versions of the churn model to identify problems.

- **Splitting The Dataset Into Training And Testing Sets:**

Once the data has been pre-processed, we need to split the dataset into training and testing sets. The training set will be used to train the model, and the testing set will be used to evaluate the performance of the model on unseen data. It is important to split the dataset in a way that is representative of the real world distribution of the data.

HOW TO OVERCOME THE CHALLENGE OF LOADING AND PREPROCESSING A CUSTOMER PREDICTION DATASET

There are a number of things that can be done to overcome the challenges of loading and preprocessing a customer prediction dataset, including:

- **Model Deployment And Maintenance:**

Once you have built and evaluated your customer churn prediction models, you need to deploy and maintain them in a way that they can provide timely and actionable insights for your business. This can be challenging, especially if you have a dynamic and changing customer base and market environment. You need to ensure that your models are integrated with your existing systems and processes, and that they can handle new and updated data. You also need to monitor and update your models regularly, and check for any changes in their performance and accuracy. To overcome this challenge, you need to establish a clear and efficient workflow for model deployment and maintenance, and use tools and platforms that can automate and simplify these tasks.

- **Carefully Consider The Specific Needs Of Your Model:**

The best way to preprocess the data will depend on the specific machine learning algorithm that you are using. It is important to carefully consider the requirements of the algorithm and to preprocess the data in a way that is compatible with the algorithm.

- **Validate The Preprocessed Data:**

It is important to validate the preprocessed data to ensure that it is in a format that can be used by the machine learning algorithm and that it is of high quality. This can be done by inspecting the data visually or by using statistical methods.

LOADING THE DATASET

- Loading the dataset using machine learning is the process of bringing the data into the machine learning environments so that it can be used to train and evaluate a model
- The specific steps involved in loading the dataset will vary depending on the machine learning library or framework that is being used. However, there are some general steps that are common to most machine learning frameworks.

a. Identify The Dataset

The first step is to identify the dataset that you want to load. This dataset may be stored in a local file, in a database, or in a cloud storage service

b. Load The Dataset

Once you have identified the dataset, you need to load it into the machine learning environment. This may involve using a built-in function in the machine learning library, or it may involve writing your own code.

c. Preprocess The Dataset

Once the dataset is loaded into the machine learning environment, you may need to preprocess it before you can start training and evaluating your model. This may involve cleaning the data, transforming the data into a suitable format, and splitting the data into training and test sets.

PROGRAM

```
from mpl_toolkits.mplot3d
```

```
import Axes3D from sklearn.preprocessing
```

```
import StandardScaler
```

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
import os
```

```
import pandas as pd
```


Loading Dataset

```
Print(os.listdir(['WA_Fn-UseC_-Telco-Customer-Churn.csv']))
```

customer_id	gender	SeniorCiti	Partner	Depender	tenure	PhoneSer	MultipleL	InternetS	OnlineSec	OnlineBac	DevicePrc	TechSupp	Streaming	Streaming	Contract	Paperless	PaymentM	MonthlyC	TotalCharJ	Churn
7590-VHV	Female	0	Yes	No	1	No	No phone	DSL	No	Yes	No	No	No	No	Month-to	Yes	Electronic	29.85	29.85	No
5575-GNV	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	Mailed ch	56.95	1889.5	No
3668-QPY	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to	Yes	Mailed ch	53.85	108.15	Yes
7795-CFO	Male	0	No	No	45	No	No phone	DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank trans	42.3	1840.75	No
9237-HQI	Female	0	No	No	2	Yes	No	Fiber opti	No	No	No	No	No	No	Month-to	Yes	Electronic	70.7	151.65	Yes
9305-CDS	Female	0	No	No	8	Yes	Yes	Fiber opti	No	No	Yes	No	Yes	Yes	Month-to	Yes	Electronic	99.65	820.5	Yes
1452-KIO	Male	0	No	Yes	22	Yes	Yes	Fiber opti	No	Yes	No	No	Yes	No	Month-to	Yes	Credit car	89.1	1949.4	No
6713-OKO	Female	0	No	No	10	No	No phone	DSL	Yes	No	No	No	No	No	Month-to	No	Mailed ch	29.75	301.9	No
7892-POO	Female	0	Yes	No	28	Yes	Yes	Fiber opti	No	No	Yes	Yes	Yes	Yes	Month-to	Yes	Electronic	104.8	3046.05	Yes
6388-TAB	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No	No	No	No	One year	No	Bank trans	56.15	3487.95	No
9763-GRS	Male	0	Yes	Yes	13	Yes	No	DSL	Yes	No	No	No	No	No	Month-to	Yes	Mailed ch	49.95	587.45	No
7469-LKB	Male	0	No	No	16	Yes	No	No	No intern	No intern	No intern	No intern	No intern	No intern	Two year	No	Credit car	18.95	326.8	No
8091-TTV	Male	0	Yes	No	58	Yes	Yes	Fiber opti	No	No	Yes	No	Yes	Yes	One year	No	Credit car	100.35	5681.1	No
0280-XJG	Male	0	No	No	49	Yes	Yes	Fiber opti	No	Yes	Yes	No	Yes	Yes	Month-to	Yes	Bank trans	103.7	5036.3	Yes
5129-JLP	Male	0	No	No	25	Yes	No	Fiber opti	Yes	No	Yes	Yes	Yes	Yes	Month-to	Yes	Electronic	105.5	2686.05	No
3655-SNQ	Female	0	Yes	Yes	69	Yes	Yes	Fiber opti	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No	Credit car	113.25	7895.15	No
8191-KWS	Female	0	No	No	52	Yes	No	No	No intern	No intern	No intern	No intern	No intern	No intern	One year	No	Mailed ch	20.65	1022.95	No
9959-WOF	Male	0	No	Yes	71	Yes	Yes	Fiber opti	Yes	No	Yes	No	Yes	Yes	Two year	No	Bank trans	106.7	7382.25	No
4190-MFL	Female	0	Yes	Yes	10	Yes	No	DSL	No	No	Yes	Yes	No	No	Month-to	No	Credit car	55.2	528.35	Yes
4183-MYF	Female	0	No	No	21	Yes	No	Fiber opti	No	Yes	Yes	No	No	Yes	Month-to	Yes	Electronic	90.05	1862.9	No

PREPROCESSING THE DATASET

- Data preprocessing is the process of cleaning, transforming, and Integrating data in order to make it ready for analysis.
- This may involve removing errors and inconsistencies, handling Missing values, transforming the data into a consistent format, and Scaling the data to a suitable range.

Prepare data for analysis

In this section, we will perform:

- Data exploration : Basic statistical analysis of the data
 - Data exploration : Explore individual features (by looking at data distribution in dataset) and cleanup data as necessary
 - Analyse relationship of the target (Churn) and input features through visualisation
- So, without further ado, let's get started!

Total number of records

```
In[1]
```

```
Print("Total number of records in training dataset:", train_data.shape)
```

```
Out[1]
```

Total number of records in training dataset: (7043, 21)

What are the features available and what are their data type?

In[2]

```
Train_data.info()
```

Out[2]

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex:      7043 entries, 0 to 7042  
Data columns (total 21 columns):  
customerID      7043 non-null object  
gender          7043 non-null object  
SeniorCitizen   7043 non-null int64  
Partner         7043 non-null object  
Dependents      7043 non-null object  
Tenure          7043 non-null int64  
PhoneService    7043 non-null object  
MultipleLines   7043 non-null object  
InternetService 7043 non-null object  
OnlineSecurity  7043 non-null object  
OnlineBackup    7043 non-null object  
DeviceProtection 7043 non-null object  
TechSupport     7043 non-null object  
StreamingTV     7043 non-null object  
StreamingMovies 7043 non-null object  
Contract        7043 non-null object  
PaperlessBilling 7043 non-null object  
PaymentMethod   7043 non-null object  
MonthlyCharges  7043 non-null float64  
TotalCharges    7043 non-null object  
Churn           7043 non-null object  
Dtypes: float64(1), int64(2), object(18)  
Memory usage: 1.1+ MB
```

Descriptive statistics of training data

In[3]

```
Train_data.describe().transpose()
```

Out[3]

	Count	mean	std	min	25%	50%	75%	max		
SeniorCitizen	7043.0	0.162147			0.368612	0.00	0.0	0.00	0.00	1.00
Tenure	7043.0	32.371149			24.559481	0.00	9.0	29.00	55.00	72.00
MonthlyCharges		7043.0	64.761692		30.090047		18.25	35.5	70.35	89.85
		118.75								

Is there any empty data in training dataset?

In[4]

```
Train_data.isnull().sum()/train_data.shape[0]
```

Out[4]

customerID	0.0
gender	0.0
SeniorCitizen	0.0
Partner	0.0
Dependents	0.0
Tenure	0.0
PhoneService	0.0
MultipleLines	0.0
InternetService	0.0
OnlineSecurity	0.0
OnlineBackup	0.0
DeviceProtection	0.0
TechSupport	0.0
StreamingTV	0.0
StreamingMovies	0.0
Contract	0.0
PaperlessBilling	0.0
PaymentMethod	0.0
MonthlyCharges	0.0
TotalCharges	0.0

Churn 0.0

Dtype: float64

Observations So Far:

There are 7043 records in the dataset.

There are 21 features in the dataset – “Churn” is the target/dependent variable (this is what we will try to predict), and rest 30 are independent variables which we need to explore further.

Only 3 features are numeric – “SeniorCitizen” (categorical), and “tenure” and “MonthlyCharges” (continuous).

Datatype of the rest of the features is object, looking at the sample data they look like to be of type string. Some of these features are categorical, which we will map into numerical values.

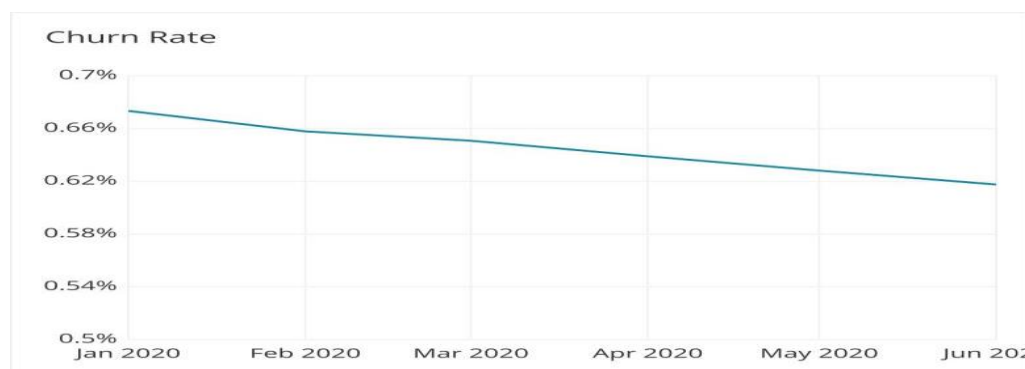
There is absolutely NO missing data. This is amazing!

VISUALIZING AND PREPROCESSING OF DATA

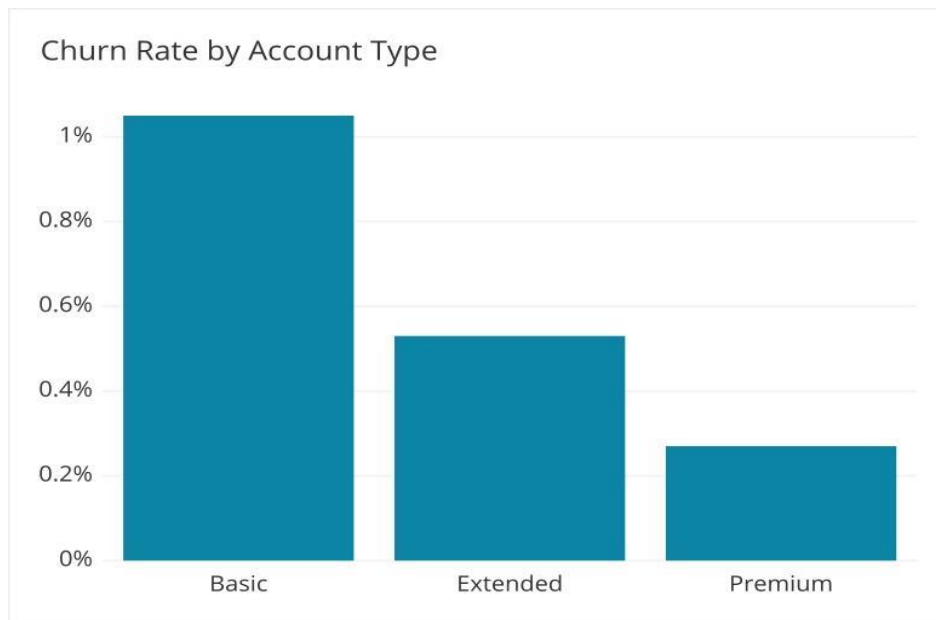
How to Visualize Churn

Data visualization allows you to quickly detect patterns, trends, and outliers in churn, enabling you to make key business decisions that can address negative patterns before they get out of control.

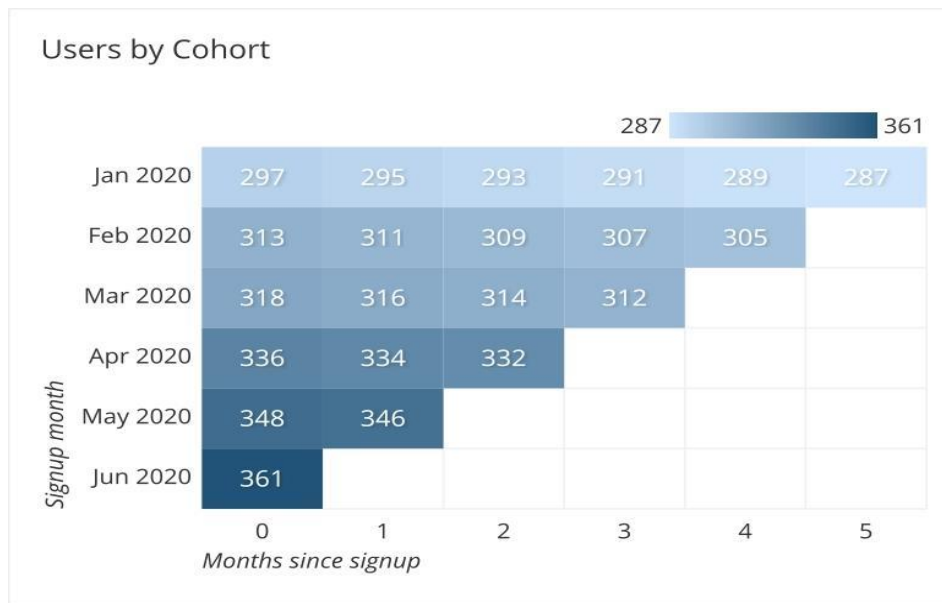
- **Single Value Charts** are great to get a quick understanding of churn over a fixed period. You can add an indicator to signal how much that rate has changed over the previous period. Single value charts can show up-to-date churn rate calculations.
- **Line charts** can be used to show churn over time. Line charts can show the churn rate trend over multiple periods



- **Bar charts** are used to show churn by different categories or can be used as an alternative to line charts to show churn over time. A bar chart can be used to compare churn across different customer segments.



- **Stacked bar charts** allow you to add another dimension to a bar chart, letting you to see in more detail where the churn was coming from. It is also common to see stacked bar charts over time to understand how these categories vary in contributing to churn.
- **Cohort analysis** is an effective way of segmenting customers to calculate key metrics, such as CLV, retention rate, and churn. Cohorts are groups of users that share a certain characteristic, such as when they signed up. A plot of churn by cohort in the form of a heatmap can help you identify when users tend to churn, enabling you to identify problems and hopefully save later customers before they reach that point.
- **A heatmap** can be used to track users by cohort and churn over time.



VISUALIZING CORRELATION

The Correlation Funnel process uses 3 functions:

- Transform the data into a binary format with `binarize()` – This step prepares semi-processed data for an optimal format (binary) for correlation analysis
- Perform correlation analysis using `correlate()` – This step correlates the “binarized” data (binary features) with the target
- Visualize the feature-target relationships using `plot_correlation_funnel()` – This step produces the visualization from which we can get business insights

Correlate to the Target

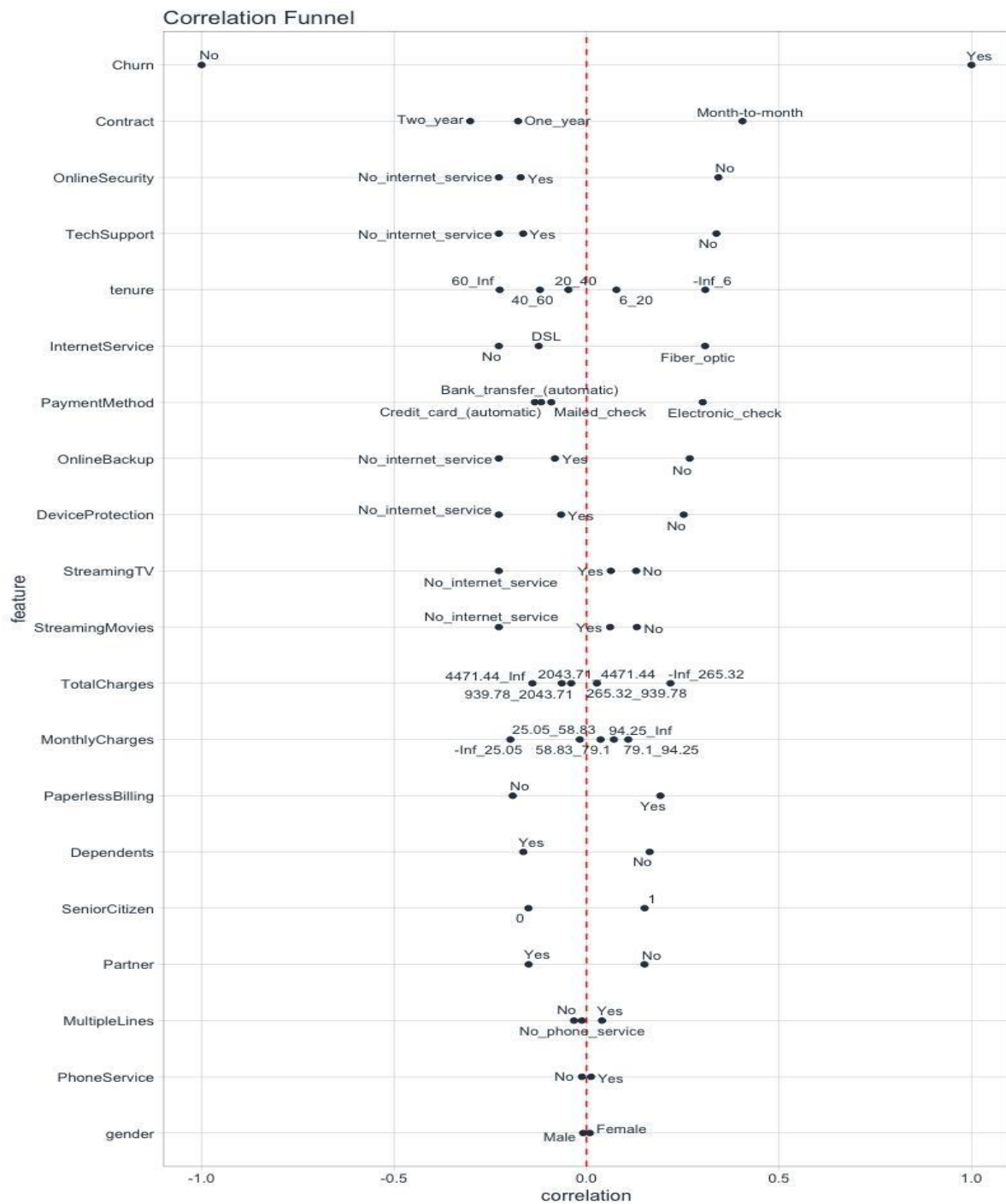
Next, we use `correlate()` to correlate the binary features to a target (in our case Customer Churn).

Plot the Correlation Funnel

Finally, we visualize the correlation using the `plot_correlation_funnel()` function.

```
Customer_churn_corr_tbl %>%
```

```
  plot_correlation_funnel()
```



COMMON DATA PREPROCESSING TASKS

Data cleaning, Data transformation, Data reduction, and Data integration are the major steps in data preprocessing.

Data Cleaning

Data cleaning, one of the major preprocessing steps in machine learning, locates and fixes errors or discrepancies in the data. From duplicates and outliers to missing numbers, it fixes them all. Methods like transformation, removal, and imputation help ML professionals perform data cleaning seamlessly.

Data Integration

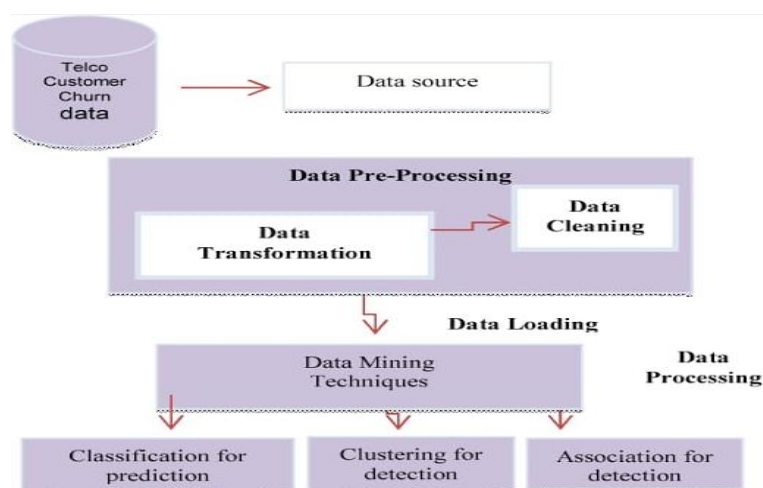
Data integration is among the major responsibilities of data preprocessing in machine learning. This process integrates (merges) information extracted from multiple sources to outline and create a single dataset. The fact that you need to handle data in multiple forms, formats, and semantics makes data integration a challenging task for many ML developers.

Data Transformation

ML programmers must pay close attention to data transformation when it comes to data preprocessing steps. This process entails putting the data in a format that will allow for analysis. Normalization, standardization, and discretisation are common data transformation procedures. While standardization transforms data to have a zero mean and unit variance, normalization scales data to a common range. Continuous data is discretized into discrete categories using this technique.

Data Reduction

Data reduction is the process of lowering the dataset's size while maintaining crucial information. Through the use of feature selection and feature extraction algorithms, data reduction can be accomplished. While feature extraction entails translating the data into a lower-dimensional space while keeping the crucial information, feature selection requires choosing a subset of pertinent characteristics from the dataset.



PROGRAM

In[1]:

```
import pandas as pd

df = pd.read_csv('WA_Fn-UseC_-Telco-Customer-Churn.csv')
```

In[2]:

```
df.head()
```

Out[2]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	

Exploratory Data Analysis for Customer Churn Prediction

In[3]:

```
import matplotlib.pyplot as plt

import seaborn as sns

import numpy as np

cols = ['gender', 'SeniorCitizen', 'Partner', 'Dependents']

numerical = cols

plt.figure(figsize=(20,4))

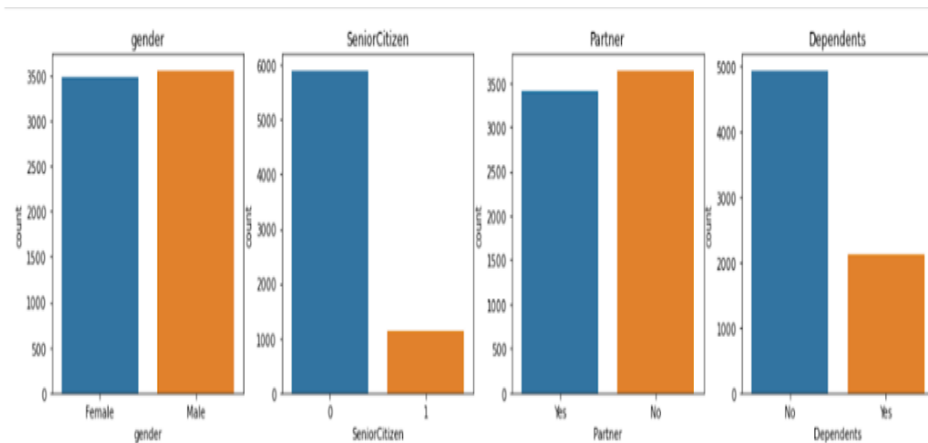
for i, col in enumerate(numerical):

    ax = plt.subplot(1, len(numerical), i+1)

    sns.countplot(x=str(col), data=df)

    ax.set_title(f'{col}')
```

Out[3]:



CONCLUSION

In conclusion, our customer churn prediction project has successfully culminated in the creation of a powerful tool that holds the potential to transform how businesses manage customer retention. Throughout this journey, we have analyzed, preprocessed, and engineered data, explored various machine learning algorithms, and rigorously tested our predictive model. The results are promising, as our model demonstrates its ability to identify customers at risk of churning with a high degree of accuracy.

By implementing this solution, businesses can proactively address churn, thereby improving customer retention rates and enhancing long-term profitability. It is important to emphasize that this project represents just the beginning of a continuous improvement cycle. To maximize its impact, ongoing data collection, model refinement, and real-time implementation are essential.

In an era where customer loyalty and satisfaction are paramount, our churn prediction model equips businesses with the insights needed to cultivate enduring customer relationships and secure a competitive edge in the market. As we look ahead, the journey continues, as we refine and enhance our solution to meet the ever-evolving needs of the business landscape. Customer churn prediction is not just a project; it's a strategic advantage for those committed to staying at the forefront of customer-centric practices and ensuring lasting success in their respective industries.